Understanding Questions: Interaction between Interviewers and Respondents

SurvMeth 632

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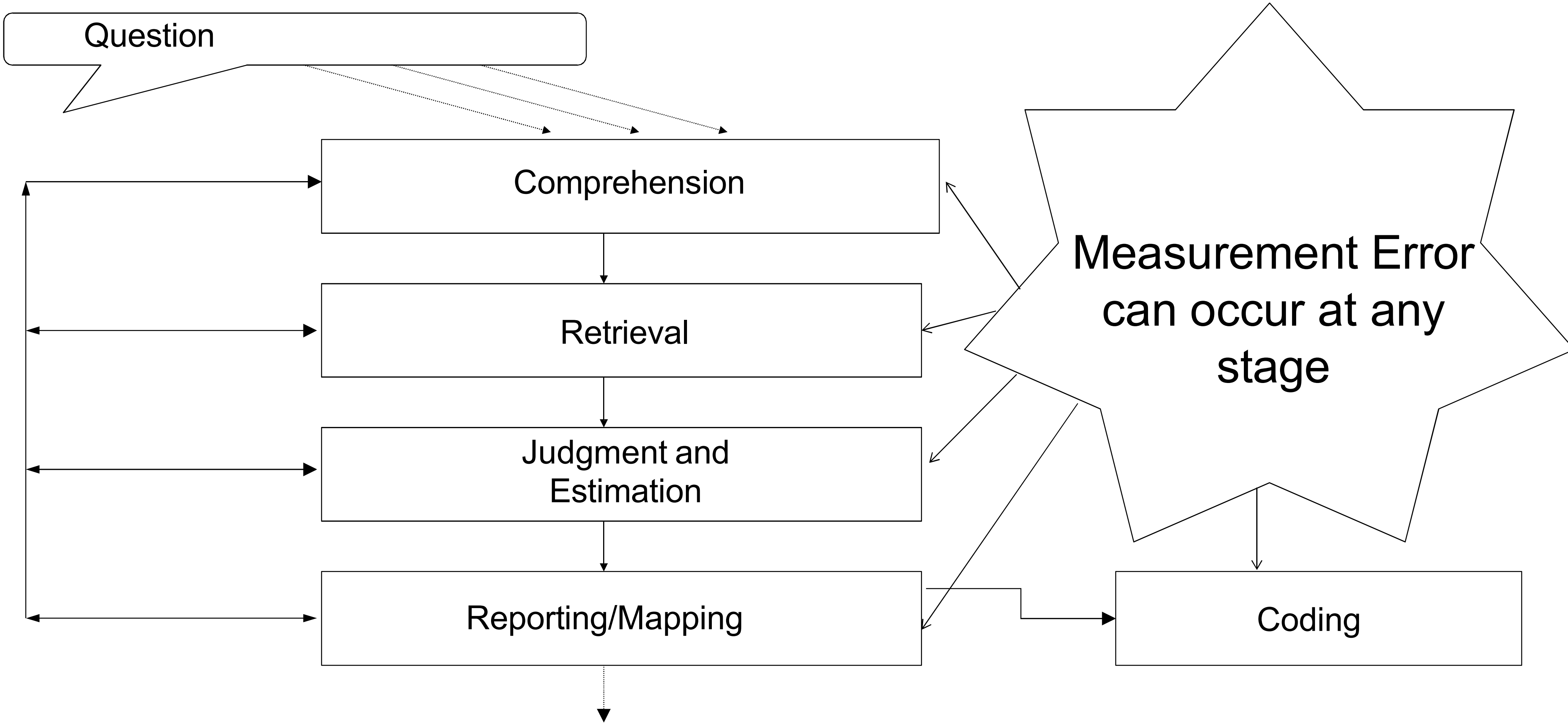
September 11, 2024

Review

1. What are the four cognitive processes that respondents use to answer survey questions?
2. What is measurement error?
3. What is the difference between the representation of the sentence and the representation about the sentence?
4. What is the cooperative principle?
5. What is common ground?
6. What is grounding?
7. What is the difference between standardized interviewing and conversational interviewing?


Question
Comprehension
Retrieval
Judgment and
Estimation
Reporting/Mapping
Measurement Error
can occur at any
stage
Coding

Literal versus non-literal meaning

- *Representation of* the sentence (question)
  - Specification of the underlying grammatical and logical structure of a sentence
  - Meanings of words (in mental dictionary) relatively stable and conventional
  - *Semantics*: word meanings and their combination
- *Representation about* the sentence (question)
  - People infer meaning beyond the (conventional) meanings of words
    - *response categories* lead *R*s to infer intensity or severity, e.g. How often are you depressed? Once a week…. Every day" vs. "Once a year ... Once a week."
    - previous *Q*s and answers may affect interpretation of current *Q*
    - *R*'s beliefs about purpose of survey may affect interpretation of *Q*
  - Inferences vary between *R*s, less conventional than word meaning
  - *Pragmatics*: language use; <u>inferences</u> about intended meaning

Complexity

- Complex questions place burden on working memory
  - During the past 12 months, since January 1, 2024, how many times have you seen or talked to a doctor or assistant about your health? Do not count any time you might have seen a doctor while you were a patient in a hospital, but count all other times you actually saw or talked to a medical doctor of any kind.
  - Some people feel the federal government should take action to reduce the inflation rate even if it means that unemployment would go up a lot. Others feel the government should take action to reduce the rate of unemployment even if it means the inflation rate would go up a lot. Where you place your self on this [7 point] scale?
- Consequences of overburdening working memory
  - Items drop out of working memory
  - Cognitive processing slows down

Literal Meaning: Conceptual Variability

- The same word means different things to different people

  Do you think children suffer any ill effects from watching programmes with violence in them, other than ordinary Westerns?

- Belson (1981) determined that *R*s interpreted *children*, *ill effects* and *violence* in numerous ways
  - “children”: < 8 years, < 19 - 20 years
  - children as students
  - only 8% interpreted all words in question as intended

❖ Caveat: We don’t know if this led to incorrect responses

- For someone who thinks *people* suffer no ill effects from watching programs with violence, misinterpreting *children* won’t affect response

Ambiguity

- Lexical

  The best way to prevent cancer is to catch it early

  strongly agree   somewhat agree   somewhat disagree   strongly disagree

  – Unclear, at least initially, which sense of "catch" is intended
    - "Become ill with" or "detect"

  – Word has more than one meaning

- Mapping

  I: Last week, did you have more than one job including part-time, evening or weekend work?

  R: Um...I babysit for two families. Is that one job or two?

  – Unclear how to apply (map) "more than one job" to one's circumstances

  – What should be *included,* and what should be *excluded*?

Beyond Literal Meaning: Presupposition

- Some sentences (questions) refer to what speaker feels listener knows or believes, either from previous conversation or just because speaker feels listener would know or believe this
- This is *presupposed*
- What if presupposition is not accepted by R?

  Family life often suffers because working parents concentrate too much on their work. Do you strongly agree, somewhat agree, etc.?

  – What is presupposed?
  – Does offering a "don't know" option solve the problem?

Beyond Literal Meaning: Common Ground

(H.H. Clark, e.g., 1996)

- Listeners go beyond literal meaning based on their knowledge of the speaker and what
  - has been said up until now
  - has been presupposed
  - the "other" can be reasonably assumed to know
  - inferences about speaker's intentions have been made
    - bridging inferences (Clark '75)

    I met a man yesterday. The bastard stole all my money.

    Alex went to a party last night. He is going to get drunk *again* tonight.
- Common ground ≈ working memory

Grice's Cooperative Principle

- Listeners infer what speaker means by assuming statements are intended to promote communication, i.e., speaker is being cooperative
  - A: *I am out of gas*
  - B: *There is a garage around the corner*
  - B infers that A is asking for information about location of gas station (otherwise, would be uncooperative to say it)
- Cooperative Principle comprised of 4 "maxims" (rules) that listeners assume speaker is following
  - Quantity (say as much as necessary but not more)
  - Quality (say what you believe to be truthful)
  - Relation (make contributions relevant to discourse)
  - Manner (be clear)
- Intentionally violating (flouting) maxims:
  - A and B both know that X has divulged a secret about A
  - A: *X is a fine friend.*

Schwarz, Strack and Mai ('91)

Correlation of Relationship Satisfaction (specific) and Life-Satisfaction (general)

Condition	***r***
Specific-general	0.67*
Specific-general, with joint lead-in	0.18
Specific-general, explicit inclusion	0.61*
Specific-general, explicit exclusion	0.20

*$p < .05$

What is standardized interviewing?

- Prevailing philosophy and practice of collecting survey data:
  - *Iwer*s expected to read question exactly as worded
    - If answer not acceptable, *I* must use *neutral probes*
  - This approach should, in principle, promote *comparable* data across *Iwer*s
    - different responses should reflect actual differences between *R*s, not differences in question wording (stimulus)
  - Standardize the stimulus so interaction is scripted
  - Statistical goal is to remove the *Iwer* as a source of error variance
    - "interviewer variance"
- Procedure/rationale articulated by Fowler and Mangione ('90)

Grounding*

(H.H. Clark, e.g., 1996)

- Speakers and listeners indicate they believe they understand each other
  - "uh huh," "I see," "got you," nodding, other gestures
- If there is not mutual understanding, addressee can seek clarification from speaker
  - "What do you mean by …?"
- Collaborative view:
  - Grasping meaning requires "back and forth" between speaker and listener; they have grounded the speaker's meaning when they agree they understand each other <u>well enough for current purposes</u>
  - Clark & Schaeffer (1987) analyzed corpus of directory assistance calls; found ~82% of exchanges seemed problem-free but remainder involved extra dialog before listener "accepted" speaker's contribution

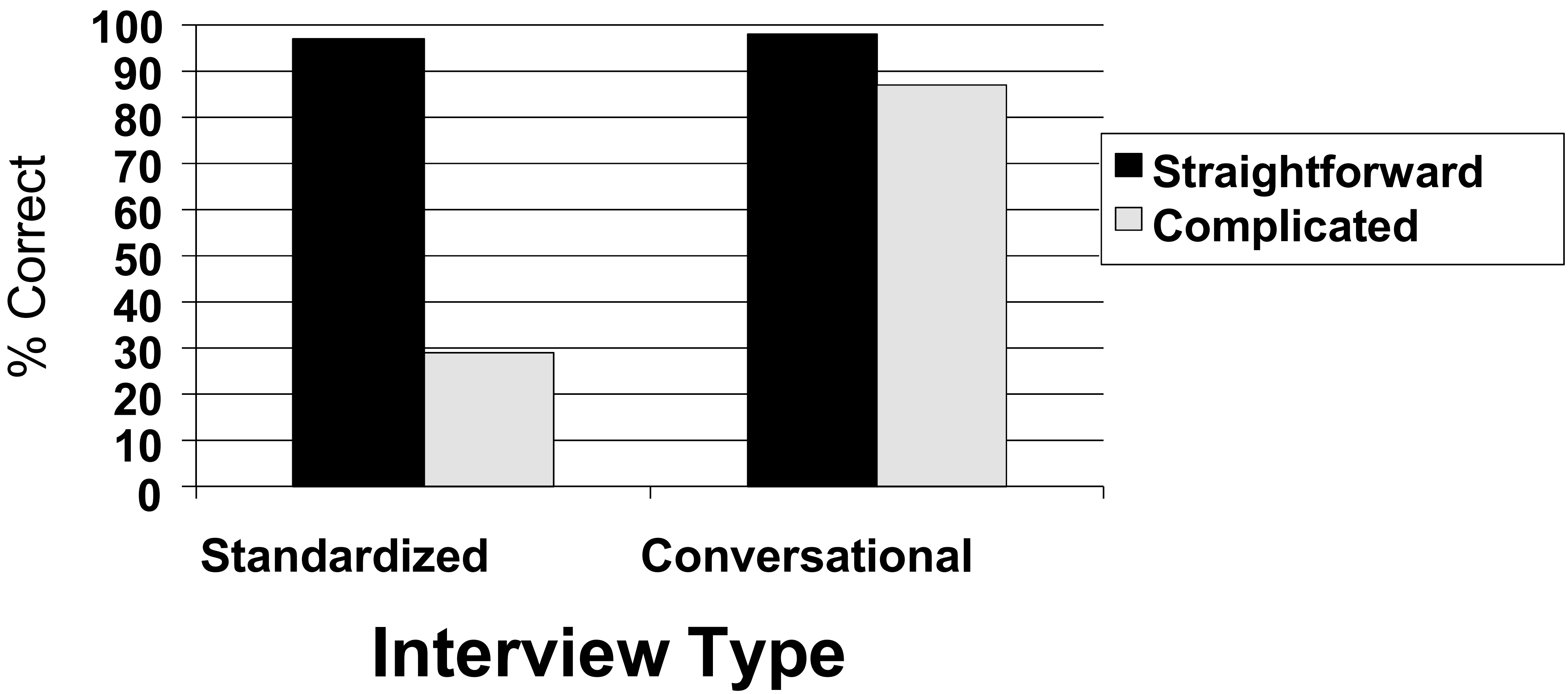
*Grounding ≠ Common Ground

Conversational Interviewing Philosophy and Procedure*

- Primary goal is to reduce mapping ambiguities
- *Iwer* authorized to say whatever it takes to be sure *R* understands *Q* as intended
  - "ground" understanding
- *Iwer* must initially read question as worded but then can clarify *Q*
  - paraphrase the question or provide a relevant definition
  - either when *R* specifically asks for clarificatio, or *Iwer* deems it necessary
- Pretesting cannot anticipate many mapping ambiguities in a diverse sample; clarification needed <u>in interview,</u> i.e., in real time when *R*s' circumstances become evident to *Iwer*

*Schober & Conrad, 1997

Response Accuracy



Minutes per Interview

	Standardized	Conversat'al
Median	3.41	11.47
Min	2.48	6.10
Max	5.99	35.44

- May overstate duration due to CI in practice because in experiment, ½ Qs involve complicated mappings; likely fewer in actual interviews

Ideal Question-Answer Sequence

- In principle, standardized interviewing leads to interactions in which *Iwer* asks a question and *R* produces an acceptable answer
  - i.e., one of the response options offered with the question
- Paradigmatic sequences (Schaeffer & Maynard, 2005)

  I: Alright and is this business or organization mainly manufacturing retail trade wholesale trade or something else (1.3)

  R: Uh, something else ← acceptable answer

  I: Alright. What kind of work does she usually do at this job, that is what is her occupation?

Departures from Paradigmatic Sequences

- May reveal interactive character of comprehension in survey interviews

  I: Is this business or organization mainly manufacturing, retail trade, wholesale trade, or something else (1.0)

  R: It's a service industry (1.8)

  I: So, it'd be under? Iwer probes "neutrally" for acceptable answer

  R: I don't know how- I don't know what you'd classify it

  I: Under something else?

  R: Yeah (1.0)

  I: And what kind of work do you usually do at this job that is (.) what is your occupation.

Source: Schaeffer & Maynard, 2005

Outline

- Granularity of *Interviewer-Respondent* interaction
  - Question
  - Turn (and position in sequence of turns)
  - Utterance (within turn)
- Cues of uncertainty and implications for response accuracy
  - Verbal and paralinguistic cues
  - Providing clarification based on cues of uncertainty
  - Visual cues, survey mode, interviewing technique

Interaction at the Question level

- May reveal problem with a question, based on a specific administration of the question
  - did *I'wer* fail to read *Q* as worded?
  - did *R* request clarification?

- When this type of analysis – called "behavior coding" -- is aggregated over many administrations of questionnaire, can identify problematic questions

- Behavior coding can provide reliable, quantitative assessment of survey response problems

Behavior Coding

1. Methodical observations of *Iwer* and *R* behaviors from audio-recorded interviews to identify likely problems in delivery and response to individual questions
2. Provides objective measures of response problems and *I*wer's ability to follow standardized interviewing is related to problems
3. Behavior coding is systematic, reliable, and relatively fast

Examples of Behavior Codes for Interviewer and Respondent Behaviors	
Code	*Description*
Interviewer Questioning Behaviors (choose one)	
1.	Reads question exactly as worded
2.	Reads questions with minor changes
3.	Reads questions so that meaning is altered
Respondent Behaviors (check as many as apply)	
1.	Interrupts question reading
2.	Asks for clarification of question
3.	Gives adequate answer
4.	Gives answer qualified about accuracy
5.	Gives answer inadequate for question
6.	Answers "don't know"
7.	Refuses to answer

Source: Groves et al. (2009)

Probes and Behavior Coding

(Oksenberg, Cannell & Kalton, 1991)

1. What was the purpose of that visit (to a health care person or organization)?
2. How much did you pay or will you have to pay out-of-pocket for your most recent visit? Do not include what insurance has paid for or will pay for. If you don't know the exact amount, please give me your best estimate.
3. When was the last time you had a general physical examination or checkup?

Percent of Problems Across All Times Question Asked

Question#	**1**	**2**	**3**
Interviewer action			
Slight wording change	2	30	3
Major wording change	3	17	2
Respondent action			
interruption	0	23	0
clarification request	2	10	3
inadequate answer	5	17	87
"don't know"	0	8	12

Behavior Coding Shortcomings

- May miss important *interactive* phenomena
  - hard to know (from the code alone) what behavior by *Iwer* preceded and may have contributed to *R*'s behavior
- May not be optimal when production schedule is tight
  - but faster than more detailed analyses
- Does not directly reveal source of problems, so results may not extend to other q'aires or help develop general principles
- Won't tell designers how to fix problems – just that problems have been identified
  - but that's true of pretesting methods in general

Turn Level Analysis

- Where in the sequence of conversational turns did the problem appear?
  - did *R* change answer after receiving a neutral probe, e.g. "Let me re-read the question?"
- Can help explain how well interviewing techniques are working or not working

Turn level analysis reveals what actually happens

(1) Adult education CATI, Christine, item 12

I. .hh uh first of all I'd like to ask
you what is the highest level of
education that you took in full-time
daytime education?
R. in full-time day- a uh basic knowledge course.
I. no, I [mean the high [est-
R. [( ) [the school
I. right, [the school,
R. [the school.
I. right, daytime school right. =
R. = no, seven years of elementary school.
I. that's- that's it?
R. yes.

Source: Houtkoup-Steenstra, 2000, pp. 91-92

Non-answer ("report") followed by "neutral probe"

I: # (0.3) During the past four weeks, have you been able to see well enough to read ordinary newsprint (0.1) without glasses or contact lenses? Y/N Q

(1.2)

R: Ah(0.1) Just reading glasses. Report

(0.7)

Neutral probe

I: Okay. I'm just going to reread that question.

R: Oh. Okay.

(0.1)

I: uh During the past four weeks, (0.1) have you been able to see well enough (0.1) to read ordinary newsprint (0.1) without (0.2) glasses (0.1) or contact lenses?

(0.3)

R: Ah (0.3) No. Acceptable answer

I: Okay?

Source: Schaeffer and Maynard , 2008, pp. 43-44

Turn-level Transcript of *I-R* interaction*

Schober & Conrad, 1997

I: Last week did Pat have more than one job, including part-time, evening or weekend work?
R: Um . s- say that again, because *[laughter]*
I: *La-*
R: She has many clients which she . but it's the same kind of job.

R reports on (describes) situation; not an acceptable answer

I: Okay. U:h *that would-*
R: *In other* words she is um .
I: Well what kind of work *does she do.*
R: *She ba-* she babysits, and she *has*
I: *O-*
R: different clients.
I: Okay, that would be considered as all one job,

I gives definition.

R: *All right*
I: *no matter* how many people she- she worked *for.*
R: *Yes* if it's the same type of job, yes, she has one job *and that's all.*

R gives acceptable answer

I: *And is this-* this is the only thing that she does.
R: Yes, and this is it.
I: Okay, so we'll say no for this. She only has one job?
R: She only has one job.

R confirms acceptable answer

I: And um . (goes on to the next question)

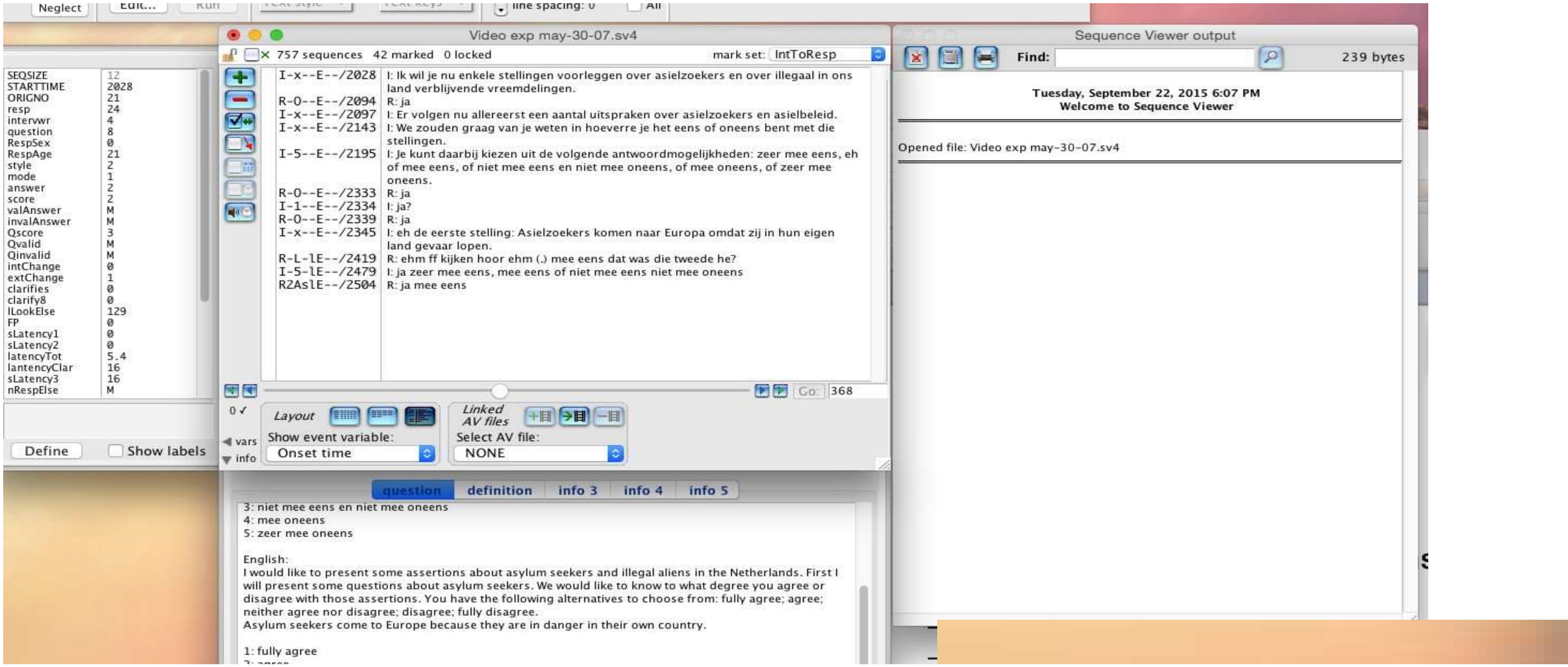
*in a conversational interview

When did conversational *I*s intervene?

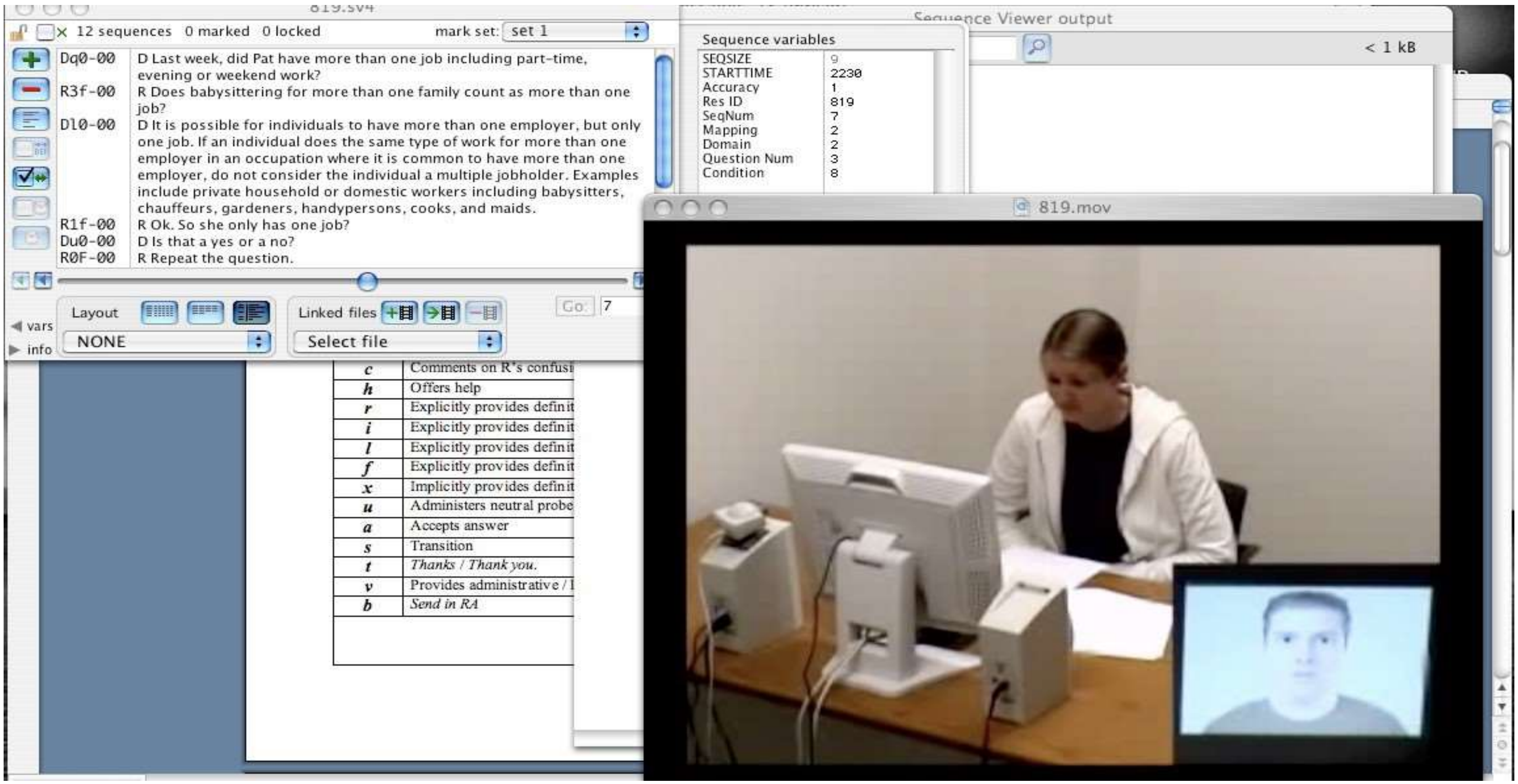
(Schober & Conrad, 1997)

R's "Move"	% Complicated Mappings*
Reports (description of situation)	37
Requests for repeat of question	9
Indication of uncertainty about answer	6
Miscellaneous	27

Sequence Viewer (SV) interface



SV Interface with Video



Utterance Level Analysis: Cues of Uncertainty

- What is speaker's cognitive or metacognitive state?
  - Did the speech include disfluencies such as *um*s and *uh*s which might indicate uncertainty?
  - Did cues of uncertainty occur more before inaccurate answers?
- Can interviewers be trained (or automated interviewing systems designed) to exploit such cues?
- *Note: "non-verbal" can mean non-lexical (as here, e.g.* uh and um*) or non-oral/auditory, i.e., visual cues (as used later)*

When listeners/respondents are uncertain about what conversational partners mean…

- they don't always explicitly ask for clarification; they may
  - Wait to hear more
  - Pretend all is well
  - Hedge and stammer
- Survey respondents answering factual questions asked for help 38% of the time, but misunderstood 70% of the time
  - Schober & Conrad (1997)
- This poses a problem for anyone who wants to be sure they have been understood
- Also, a problem for researchers who want to be sure participants interpret questions or instructions as intended

Cues of Uncertainty When Misunderstanding is Likely

- Schober & Bloom (2004) identified cues that researchers or interviewers can (potentially) exploit to determine if *R*s who don't explicitly ask for help are at risk of misunderstanding
- Analyzed Schober & Conrad (1997) corpus
  - Because true value known (answers based on scenarios), possible to determine if *R*s answer accurately or not, and thus if cues predict comprehension problems

Potential cues of uncertainty: 1. Speech disfluencies

- Pauses (mid-clause or pre-utterance)
  - "Last week I bought (*1.2 s*) a lamp."
  - "(*2.3 s*) I bought a lamp."
- Filled pauses ("fillers")
  - "Last week I bought a *um* lamp."
- Repairs
  - "L- l- last week I bought a fl- lamp."

2. Hedges

- *"*We have *about* four bedrooms.*"*
- *"I think* we have four bedrooms.*"*

3. Discourse markers

- May alert addressee that what comes next is unexpected or that speaker isn't entirely sure (Schiffrin, 1988)
  - "Well, I bought a lamp."

4. Reports

- Answering a question in a way that leaves responsibility of answering to questioner (Drew, 1985)
  - A: Do you like punk rock?
  - B: I like the Clash.

Explicit* Requests for Clarification

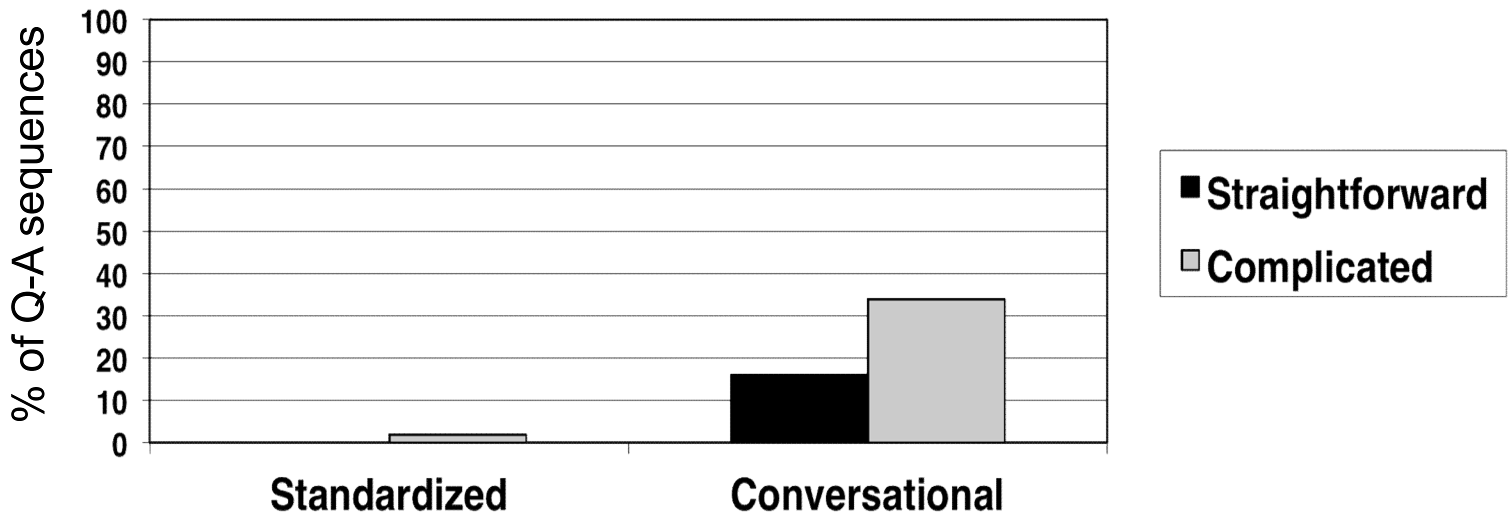
I: How many people live in this house?

R: Full- time?

- Does *R* request clarification immediately after *Iwer* asks *Q*?

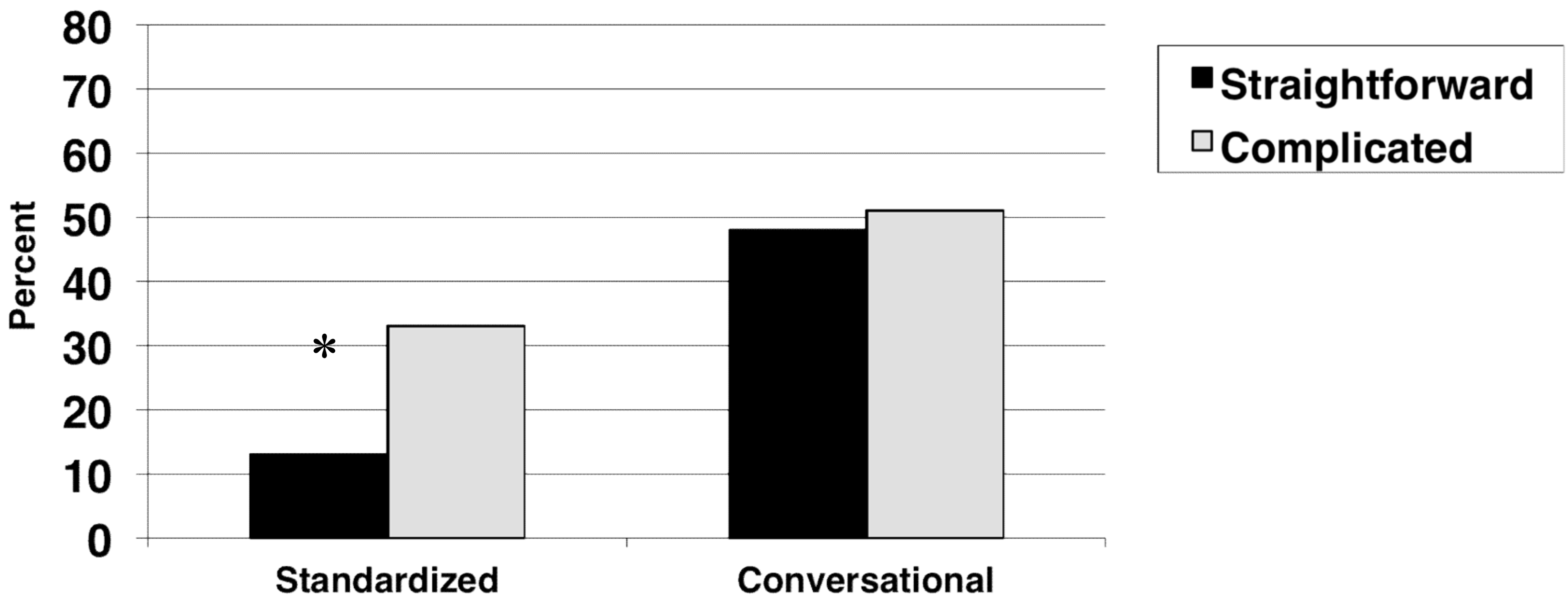
*broadly defined

Explicit requests for clarification immediately after I asks question



Schober and Bloom, 2004

*R*s' fillers immediately after *I* asks question



* $p < .05$

Schober and Bloom, 2004

Uncertainty cues can be useful for researchers

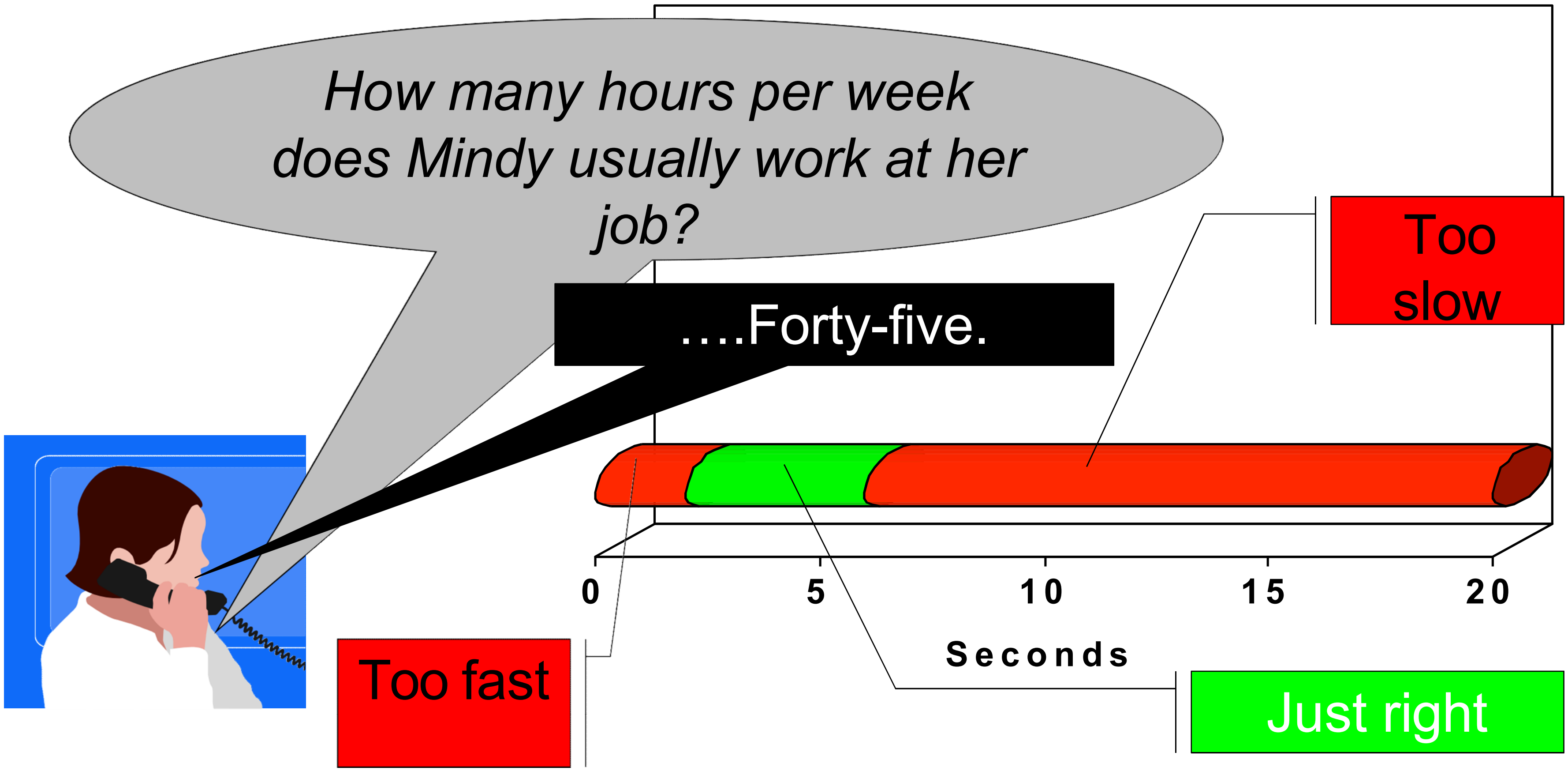
- because some uncertainty cues predict complicated situations (mappings), which may lead to inaccurate answers
- *I*s might be trained to respond to cues
  - If this skill is trainable
- Speech IVR systems (automated interviewing systems that recognize *R*'s speech) could be designed to use such cues

Automated Clarification and Uncertainty Cues

- Ehlen et al. (2007) predicted response accuracy* in speech IVR system on basis of uncertainty cues
  - *Um*s and *uh*s, hedges, restarts, repeats, repairs, reports, mutters, confirmation pick-ups and pauses
  - All of the cues predicted response accuracy, but …
- None of the cues improved the fit beyond <u>pause length</u>:
  - Accuracy poor if response *too slow or too fast* compared to “just right” zone

* using scenarios

Pause before Responding as Indication of Difficulty



Mode and Ability to Exploit Cues

- *R*s who are uncertain but do not explicitly state their uncertainty may provide implicit cues of their uncertainty
- How might this be affected by mode of communication?
  - e.g., on the telephone, *Iwer* cannot see *R*'s face but in in-person *Iwer* can take *R*'s facial expressions into account
- How might this be affected by *Iwer*'s ability to provide clarification in response to cues?
  - In standardized interviews, *I*s cannot provide explicit clarification

Schober, Conrad, Dijkstra & Ongena (2012)

- 2 Interviewing Techniques:
  - Standardized
  - Conversational
- 2 Modes
  - Face to Face (FTF)
  - Telephone

	Stand.	Conv.
FTF	11	10
Telephone	11	10

Number of Interviews

- Post-interview Q'aire with definitions
  - Change between original and subsequent answers treated as "measure" response accuracy
  - More change indicates original misconception

Visual Cues of Comprehension* Trouble

- Facial expression
- Head movement
- Gaze aversion
  - e.g., Glenberg, Schroeder & Robinson (1998)
- If cues displayed to convey uncertainty, should be more frequent in conversational interviews, where *I* can act on basis of cues

*or, response trouble, more generally

Results

Verbal Cues: Fillers

- Rs produce more fillers (per 100 words) on phone (12.3) than FTF (6.4)
  - Perhaps to compensate for lack visual cues on phone
- *R*s produce fillers in more answers in conversational (55.4%) than standardized (43.8%) interviews
  - Perhaps because they know conversational *Iwer*s can help based on cues
- *R*s more likely to change their initial answer if contained filler (9.8% vs. 2.1%)
  - fillers can reflect comprehension difficulty (e.g., Schober & Bloom, 2004)

Visual Cues: Gaze Aversion

- *R*s averted gaze (at least once) in more conversational (24.7%) than standardized (11.4%) interviews
  - Perhaps because they know conversational *Iwer*s can help based on cues
- *R*s more likely to change their initial answer if averted gaze while answering (24.7% vs. 4.3%)
  - Presumably gaze aversion reflects comprehension difficulty (or response difficulty more generally)

Implications for Interviewing

- *Iwer*s could, in principle, be trained to offer clarification when *R*s exhibit verbal or visual signs of uncertainty, but some may be better suited for this than others
  - Hubbard et al. (2020) showed *Iwer*s low in nonverbal sensitivity more often provided clarification indiscriminately; *Iwer*s high in nonverbal sensitivity clarified more selectively
- Automated detection of cues
  - Speech dialog systems might be designed to detect some cues that *Iwer* might have missed
  - And alert *Iwer* to possible comprehension trouble

Summary

- There is a story about comprehension in the details of I-R interaction
  - Evident in question level approaches like behavior coding
  - Turn-based analyses provide more detail about sequential dependencies – what follows what -- and this can help reveal origin of comprehension difficulty
- When *R*s display *c*ues of uncertainty may predict comprehension difficulty and increased likelihood of inaccurate responses
- *R*s may produce cues differently depending on whether and how *Iwer*s can detect and react to them
  - *R*s produce more verbal cues when visual cues will not be detected, i.e., on the phone, and more cues when *Iwer* can help in response to cues, i.e., in conversational than standardized interviews
- Automated detection of *R*s' cues of difficulty may potentially support/augment *Iwer*s' detection